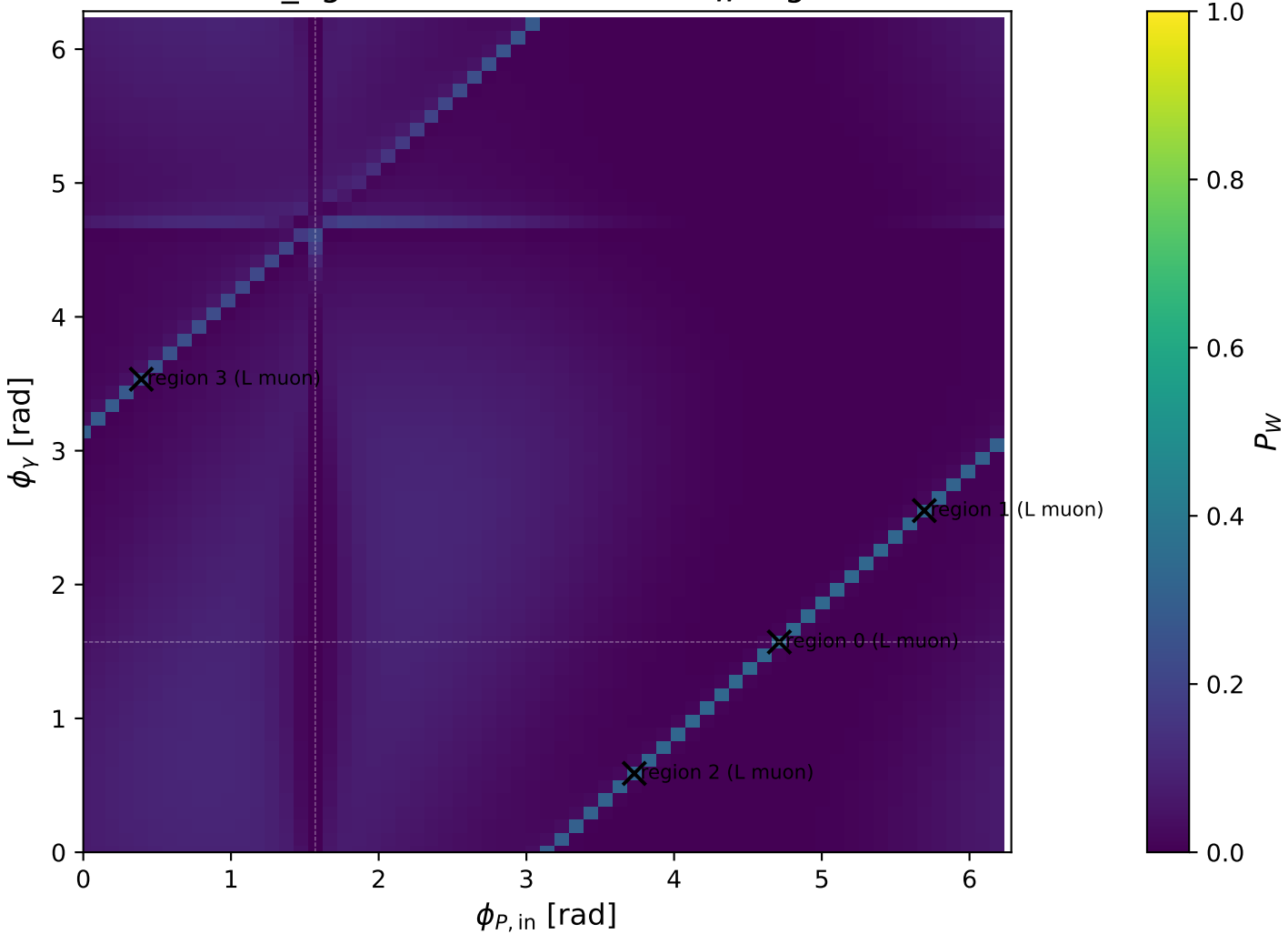
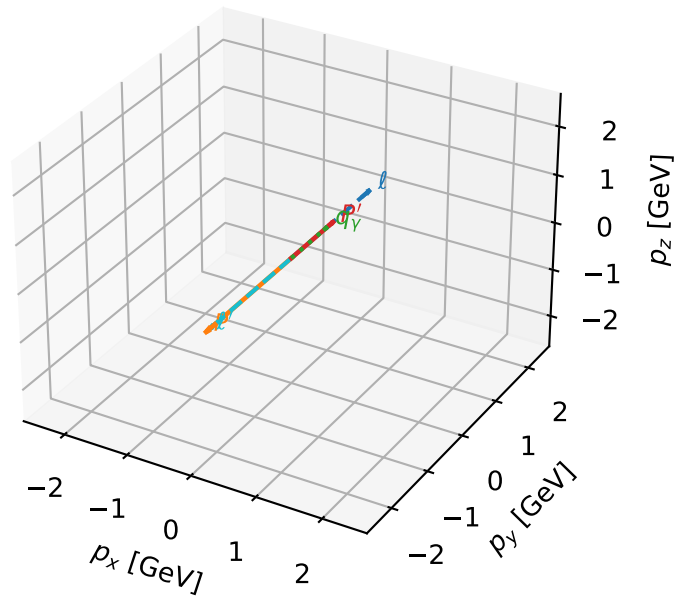


mid_Egamma L muon: max P_W regions



L muon characteristic max P_W configuration

3D momenta

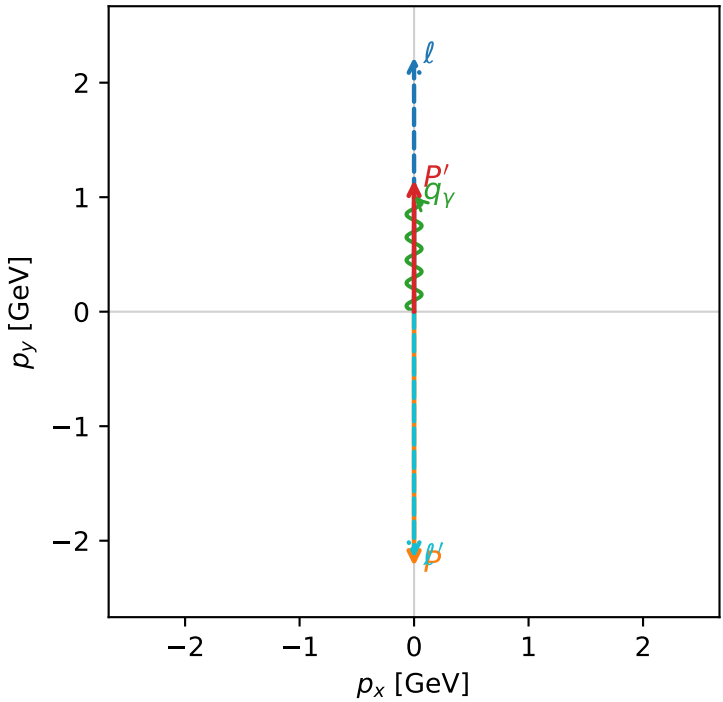


w_purity_mid_egamma_l_lepton_region_0 P_W=0.333333
region: mid_Egamma_L_lepton_region_0
outgoing-state purity: 0.999999
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_\mu, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.15107$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.5708$, $\phi_P=4.71239$
 $E_Y=1$, $\phi_Y=1.5708$
 $Q^2=19.1283$, $x_B=0.966277$, $t=-10.5238$, $W^2=1.54742$, $y=0.959806$
 $\theta(\ell', \gamma)=180$ deg

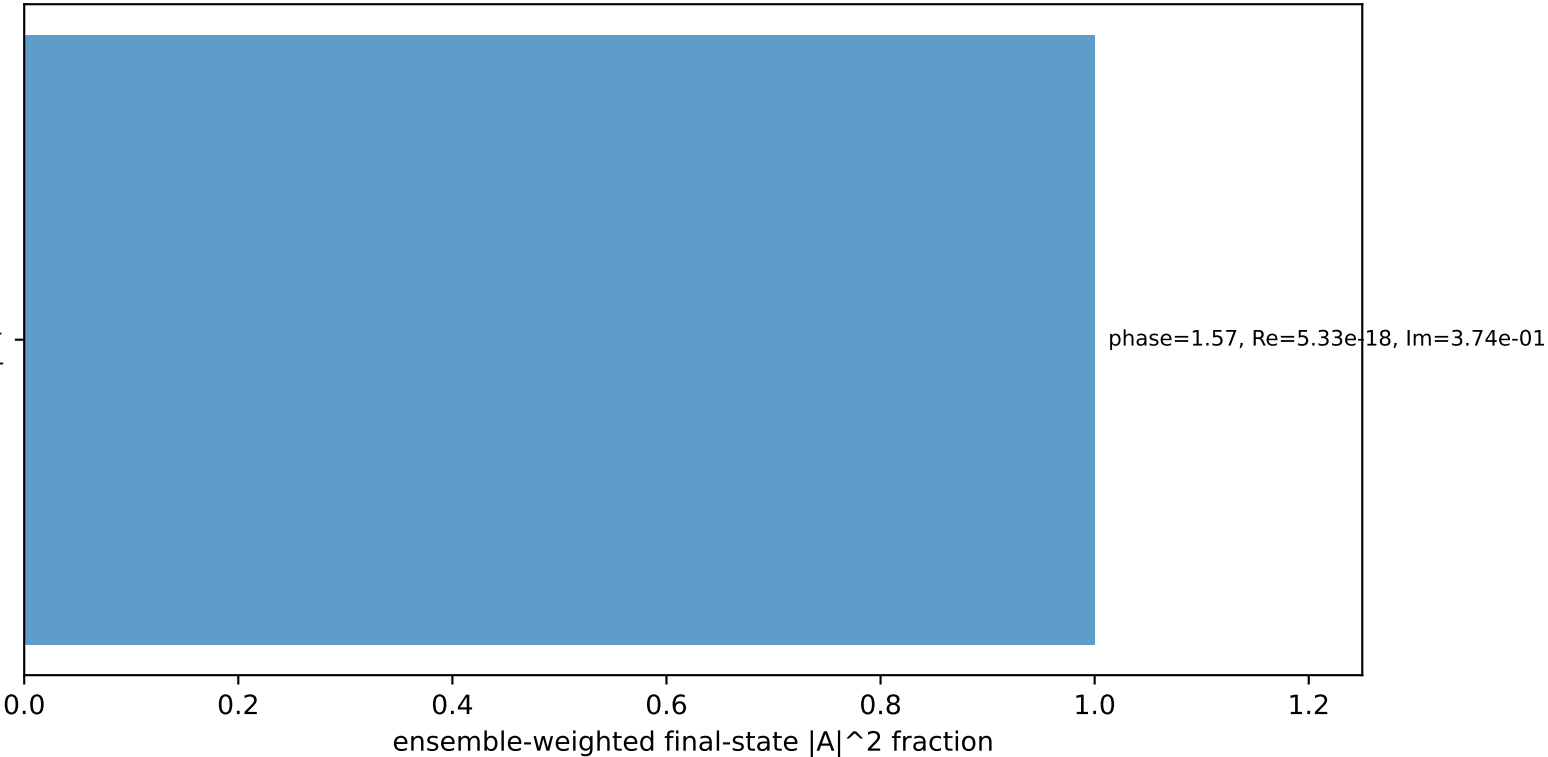
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 ℓ' $E= 2.1537$ $p_x= -0.0000$ $p_y= -2.1511$ $p_z= 0.0000$
 P' $E= 1.4849$ $p_x= 0.0000$ $p_y= 1.1511$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

Transverse plane



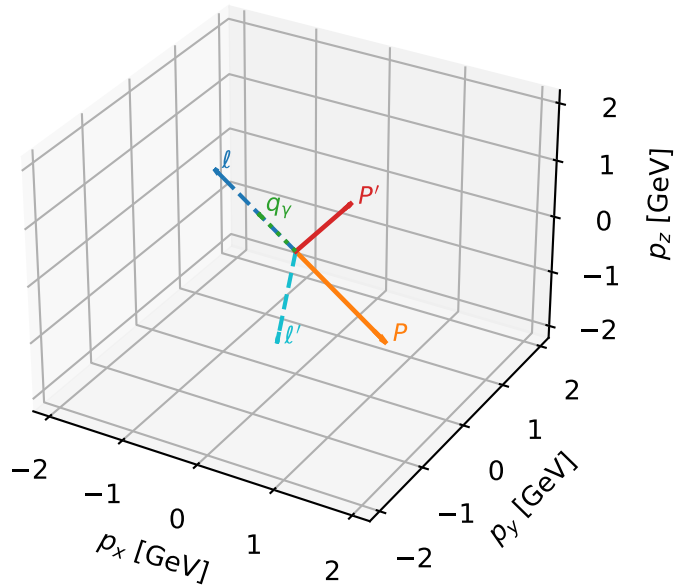
$h_e=-1, h_p=-1, h_Y=+1$
electron L, proton h=-1

Leading initial-ensemble/final-state components (N=1, retained=100.0%)



L muon characteristic max P_W configuration

3D momenta

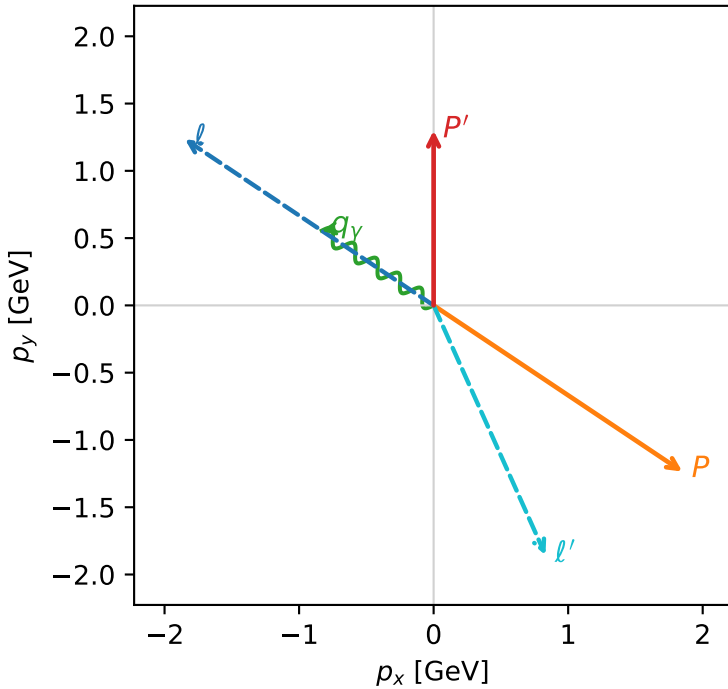


w_purity_mid_egamma_l_lepton_region_1 P_W=0.32235
 region: mid_Egamma_L_lepton_region_1
 outgoing-state purity: 0.975074
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.29963$
 $\theta_{in}=1.57079$, $\phi_l=2.55254$, $\phi_P=5.69414$
 $E_\gamma=1$, $\phi_\gamma=2.55254$
 $Q^2=16.6958$, $x_B=0.904565$, $t=-9.18528$, $W^2=2.64132$, $y=0.894906$
 $\theta(l', \gamma)=147.891$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.0357$ $p_x= 0.8315$ $p_y= -1.8552$ $p_z= 0.0000$
 P' $E= 1.6028$ $p_x= 0.0000$ $p_y= 1.2996$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

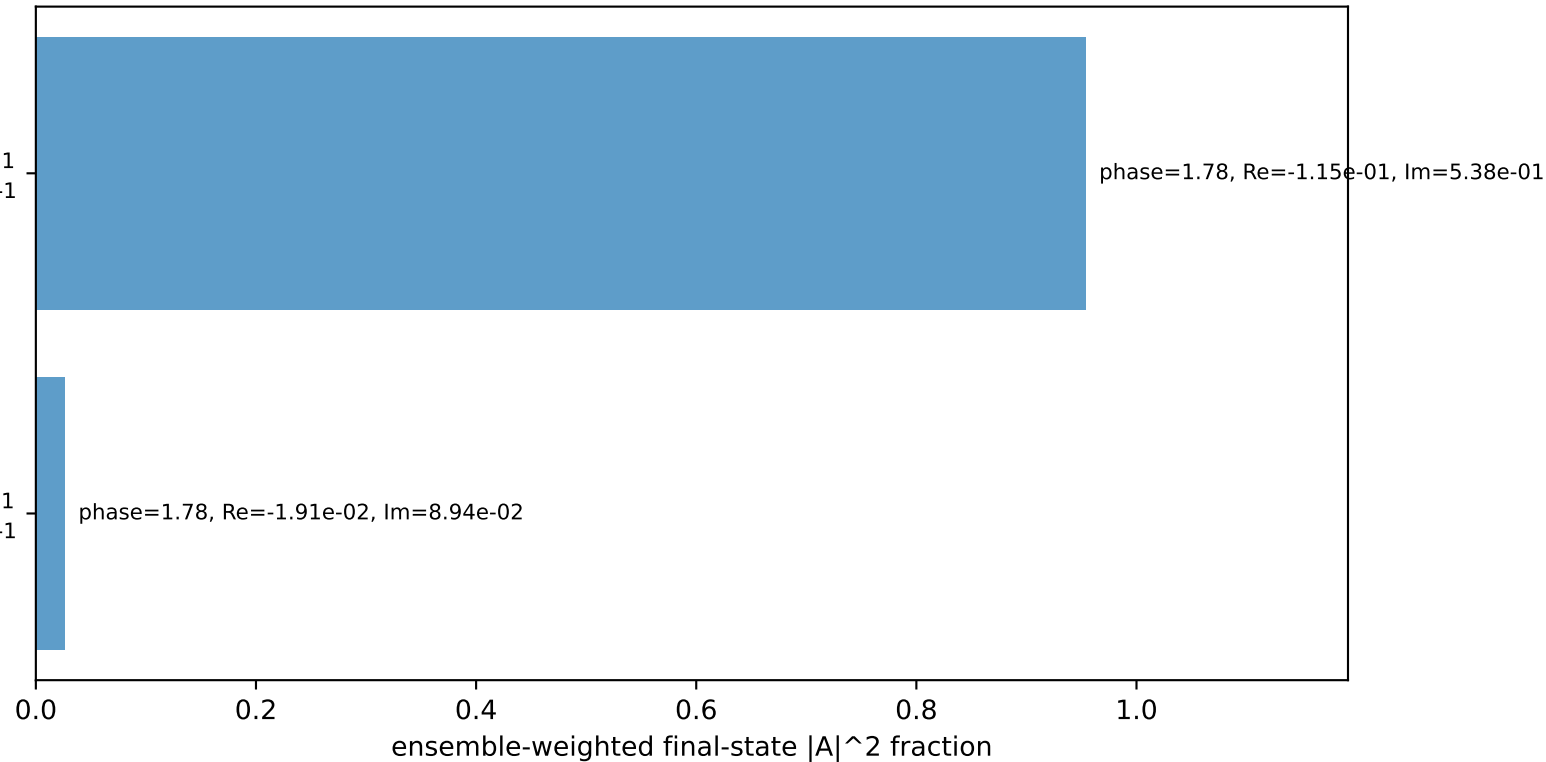
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 electron L, proton h=-1

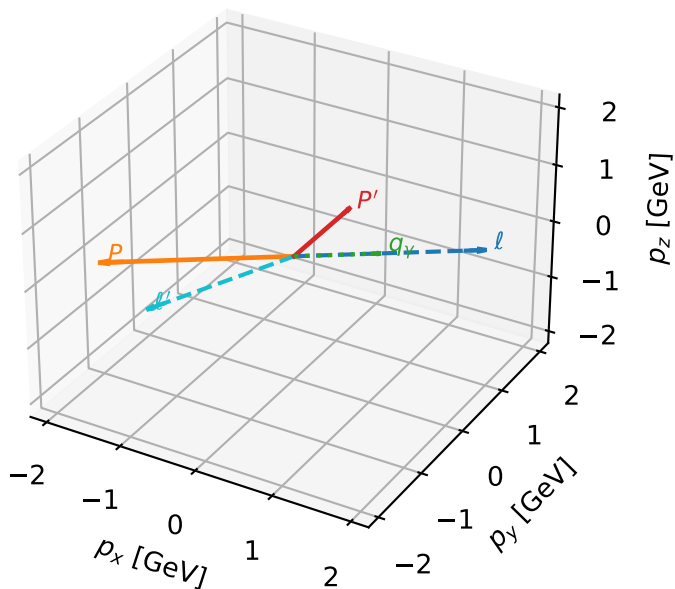
$h_e=-1, h_p=+1, h_\gamma=+1$
 electron L, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=98.0%)



L muon characteristic max P_W configuration

3D momenta

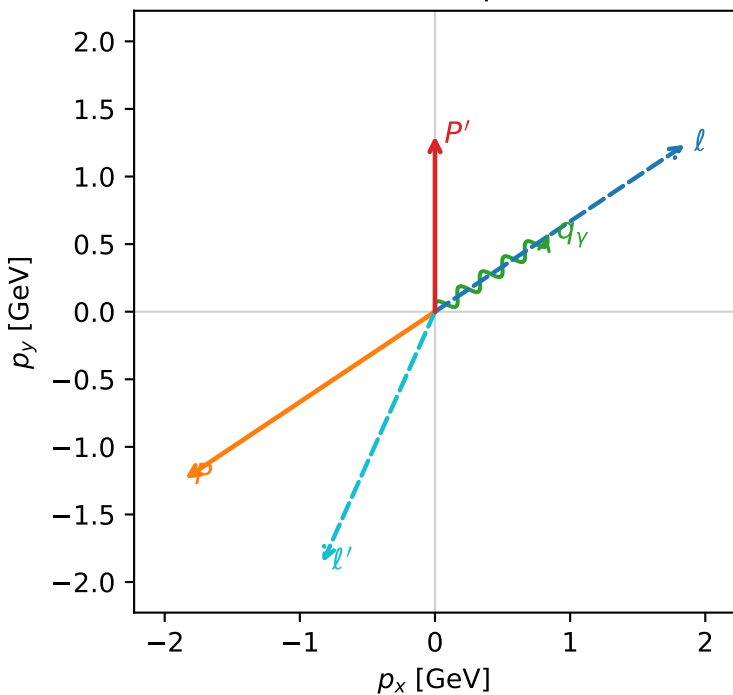


w_purity_mid_egamma_l_lepton_region_2 P_W=0.320332
region: mid_Egamma_L_lepton_region_2
outgoing-state purity: 0.975074
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.29963$
 $\theta_{in}=1.57079$, $\phi_l=0.589049$, $\phi_p=3.73064$
 $E_\gamma=1$, $\phi_\gamma=0.589049$
 $Q^2=16.6958$, $x_B=0.904565$, $t=-9.18528$, $W^2=2.64132$, $y=0.894906$
 $\theta(l', \gamma)=147.891$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.0357$ $p_x= -0.8315$ $p_y= -1.8552$ $p_z= 0.0000$
 P' $E= 1.6028$ $p_x= 0.0000$ $p_y= 1.2996$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

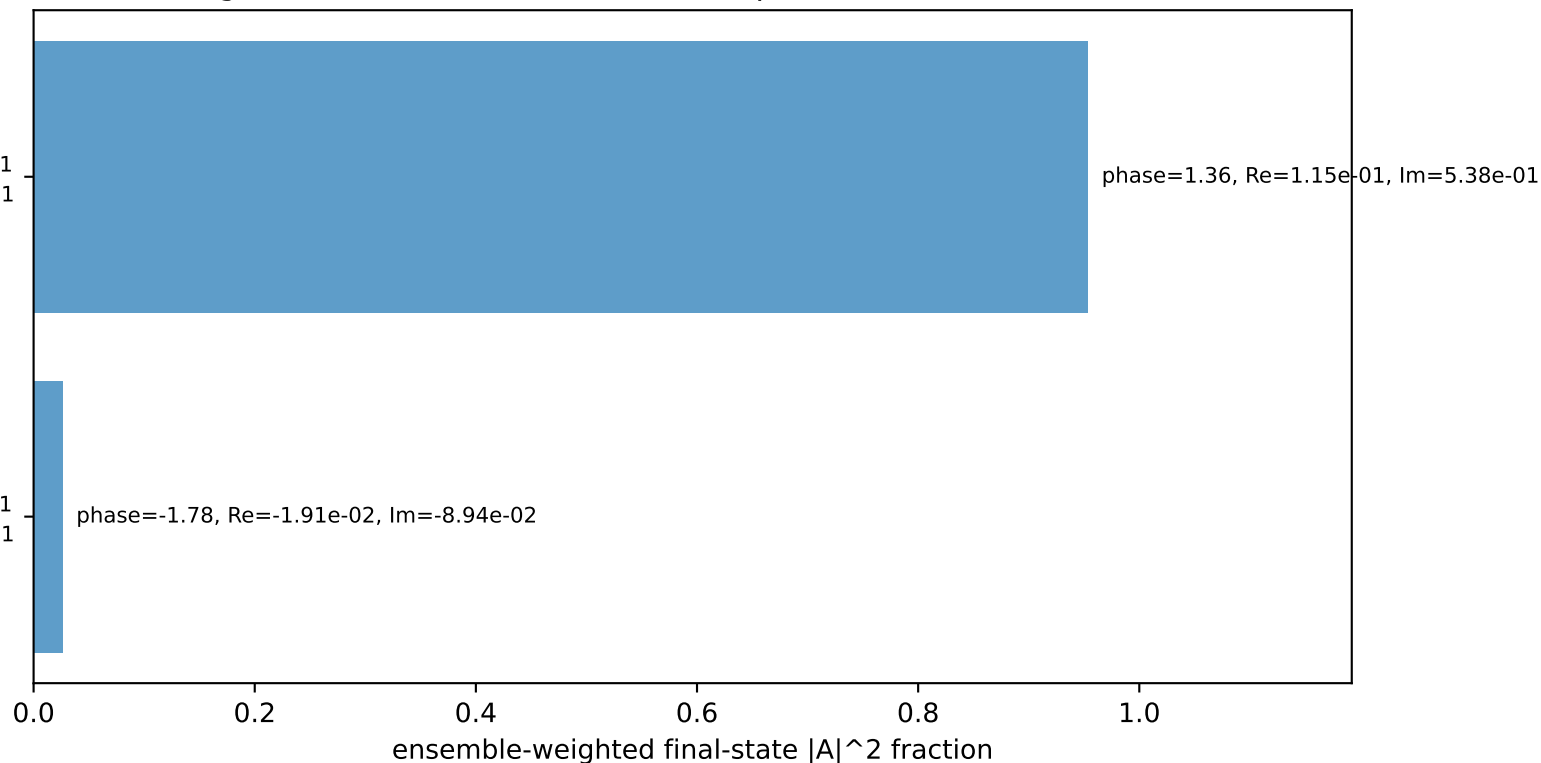
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
electron L, proton h=-1

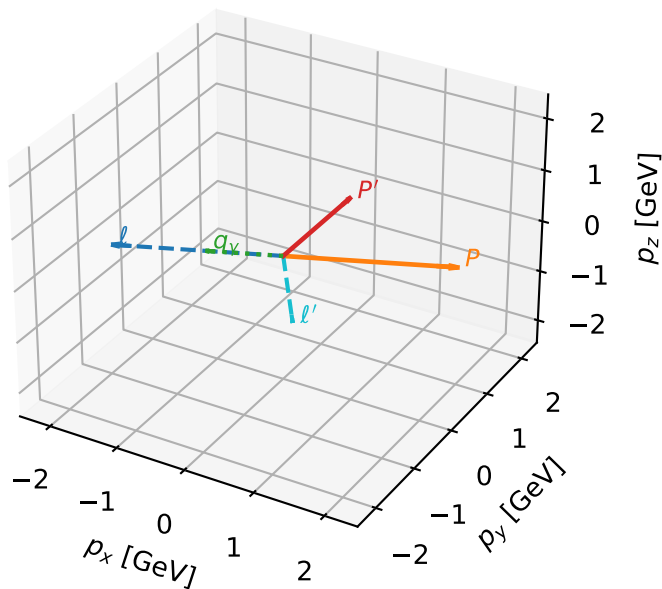
$h_e=-1, h_p=+1, h_\gamma=+1$
electron L, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=98.0%)



L muon characteristic max P_W configuration

3D momenta



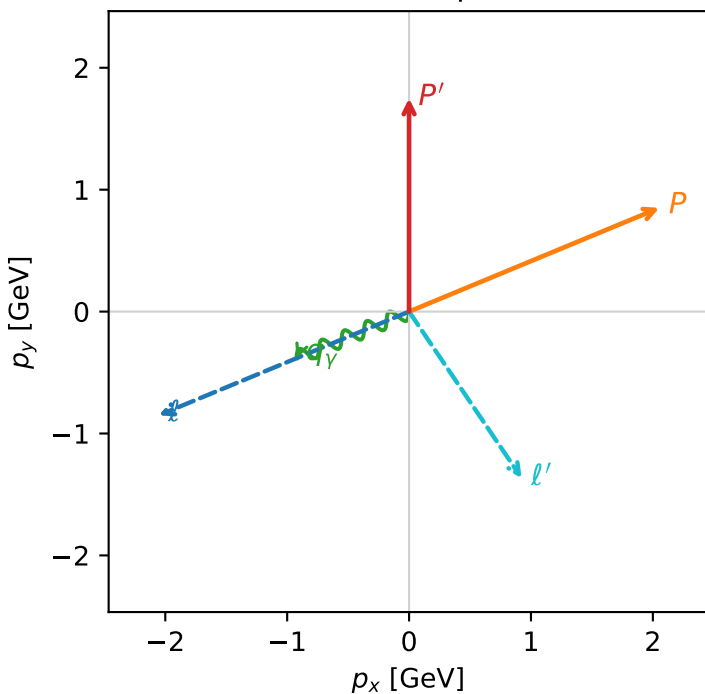
w_purity_mid_egamma_l_lepton_region_3 $P_W=0.265266$
 region: mid_Egamma_L_lepton_region_3
 outgoing-state purity: 0.694562
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_\mu, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.74966$
 $\theta_{in}=1.57079$, $\phi_l=3.53429$, $\phi_p=0.392699$
 $E_\gamma=1$, $\phi_\gamma=3.53429$
 $Q^2=8.80602$, $x_B=0.623849$, $t=-4.8436$, $W^2=6.18945$, $y=0.684399$
 $\theta(l', \gamma)=101.553$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= -2.0539$ $p_y= -0.8507$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 2.0539$ $p_y= 0.8507$ $p_z= 0.0000$
 l' $E= 1.6533$ $p_x= 0.9239$ $p_y= -1.3670$ $p_z= 0.0000$
 P' $E= 1.9852$ $p_x= 0.0000$ $p_y= 1.7497$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9239$ $p_y= -0.3827$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.4%)

